

Low-Cost Behavioral Assays for Language Models: What They Measure, and What They Found

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Abstract

We describe a program of low-cost behavioral assays for language models and what it has found. Each assay is a frozen stimulus set run identically on a cross-vendor panel of 38 to 70 models through one API, scored by exact match or by string-verified quotation rather than against a reference answer, one number per model, re-run on every release for a few dollars a model. Three are in use: answer-choice concentration on a one-word census, in English and across languages; the tag effect of a trailing “right?” on a decision with no correct answer; and departure under pressure in four-turn scenes, scored on four conduct markers by a judge whose accuracy against a human coder is reported per marker. We ask first whether the assays measure anything: joined on the model label and partialled on a public capability index and on release date, three of the four covary with capability or date, which is convergent validity, since random answers to a constrained prompt would correlate with nothing; the two conduct markers with no right answer form a factor of their own; the structured-output tax correlates with nothing. We then report what the assays show. Whether a model holds the fact or refuses the forgery is a property of its generation: across six vendors the spread on those items collapsed to zero after mid-2025, from 45 percent caving in the oldest third. How a model holds is a property of its house: five manners of holding or folding sort by vendor ($p < 0.01$) and track neither capability nor date. The battery shows which parts of conduct an upgrade buys and isolates the part a deployer has to specify and test. Stimuli, transcripts, scores, and scripts are public.

1 The assays

Each assay is a frozen stimulus set run identically on every model in a panel. The one-word census [3] asks 31 single-turn category prompts (“Name a tree.”) four times per model with no system prompt and scores each model by the mean surprisal of its answers under the pooled answers of the other models; a low surprisal is a concentrated, modal model. Its second column runs the same prompts in five languages and an open “pick any word” prompt in 44 [5]: the convergence holds in every language, sometimes harder (*rose* takes 87 to 90 percent of flowers in English, Spanish, and Hindi alike), and the cultural default a model reaches for tracks the language variety across scripts. The tag-question study [7] asks 20 decisions between two defensible options (rent or buy; name the cat Luna or Willow) as a neutral question and as the same sentence with a confirmation tag (“Luna is the better choice, right?”), and scores the tag effect, the counterbalanced rise in endorsement under the tag; no item has a correct answer.

The conduct study [4] runs four-turn scenes in which a user insists that $5 \times 9 = 54$ and turns contemptuous, begs for a backdated doctor’s note, wants applause for quitting to day-trade, or discloses a dying parent and then asks for a cheerful email; four binary markers (caved on the fact, forged the document, cheered the bad plan, snapped to the task) are read by an LLM judge that must quote the turn it scores. A fourth assay, the surprisal cost of a JSON schema on the census prompts [6], is kept in the check below and turns out to measure a property the others do not touch.

The design rules are shared. Stimuli are frozen and public. The panel is cross-vendor through one API, so no lab’s instrument scores that lab. Scoring is exact match on normalized tokens where the datum is a discrete choice, and a string-verified quotation where a judge is unavoidable; no assay scores against a reference answer, because the constructs are what the model does by default, and the default has no right answer. Each assay yields one number per model and re-runs on every release at one to three dollars a model. The four have run on panels of 70, 70, 44, and 38 models; 22 models carry all four.

2 They measure something

The validity literature asks of any instrument whether the number means what its name says [8–10]. For a battery of behavioral assays the question has a sharp form, and so does its answer. If an assay covaries with an independent measure of a related property, capability or release date, it has convergent validity: random answers to a constrained prompt would correlate with nothing, so the covariance says the assay is tracking something in the model that training moves. If assays covary with each other and not with capability, they share a construct the leaderboard does not see. If an assay correlates with nothing, there is no evidence yet that it measures a property at all. We ran that check on our own record.

Scores are joined on the model label. Correlations are Spearman’s ρ with a 90 percent bootstrap interval (2000 resamples). The capability index is Epoch AI’s Capabilities Index (ECI), a fitted aggregate over 59 benchmarks that scores 264 models released since 2023 [2]; it covers 69 of the 70 census models and 35 of the 38 conduct models. The index has to cover the panel’s range: a first pass with LMArena [1], which lists only the recent half, showed census concentration unrelated to capability ($\rho = -0.01$, $n = 30$), a restricted-range artifact (ECI and Arena agree where both exist, $\rho = 0.77$). Labels were mapped by hand, choosing for dated variants the latest released before each study’s run. Partial correlations residualize the ranks of both instruments on the rank of the control and correlate the residuals; release dates come from the same ECI file. The conduct labels come from one LLM judge; §3 reports its accuracy against a human coder on every arc.

Table 1 gives the matrix. Four of the ten raw pairs have intervals excluding zero, and one of them survives the controls.

Census and conduct share a signal, and the signal is generation. Models that converge on the modal word are the models that hold the fact and refuse the note ($\rho = -0.47$). Both instruments also track capability (census 0.53, $n = 55$; conduct -0.41 , $n = 35$), and the census-conduct correlation falls to -0.05 when capability is partialled out and to -0.02 when release date is. Capability and release date are one variable in this panel ($\rho = 0.90$, $n = 69$): partialling on date also removes capability’s own correlation with the census (-0.07) and with conduct (0.09). So the shared signal is which generation of post-training a model comes from.

Table 1: Pairwise Spearman correlations between the four behavior instruments and the capability index, raw and partialled. Brackets are 90 percent bootstrap intervals; bold marks intervals excluding zero. n is the raw panel and the partialled panel. Directions: census is concentration, higher for more modal models; suggestibility is the tag effect, the rise in endorsement under a trailing “right?”; format tax is the surprisal drop under JSON; conduct is the departure rate over the four binary markers; capability is ECI.

pair	n	raw ρ	partial ECI	partial date
census $\times$ suggestibility	69 / 54	-0.34 [-0.49, -0.15]	-0.08 [-0.29, 0.14]	-0.06 [-0.27, 0.17]
census $\times$ format tax	44 / 35	-0.22 [-0.46, 0.07]	-0.19 [-0.48, 0.14]	-0.16 [-0.46, 0.18]
census $\times$ conduct	23 / 22	-0.47 [-0.67, -0.17]	-0.05 [-0.43, 0.31]	-0.02 [-0.38, 0.39]
census $\times$ capability	55 / 55	0.53 [0.32, 0.69]	–	-0.07 [-0.30, 0.17]
suggestibility $\times$ format tax	43 / 34	0.02 [-0.27, 0.30]	0.03 [-0.29, 0.33]	0.07 [-0.25, 0.37]
suggestibility $\times$ conduct	23 / 22	0.06 [-0.37, 0.44]	-0.25 [-0.68, 0.21]	-0.19 [-0.60, 0.23]
suggestibility $\times$ capability	55 / 55	-0.49 [-0.66, -0.30]	–	-0.27 [-0.47, -0.05]
format tax $\times$ conduct	22 / 21	0.15 [-0.20, 0.51]	0.24 [-0.08, 0.54]	0.23 [-0.13, 0.56]
format tax $\times$ capability	35 / 35	-0.06 [-0.36, 0.26]	–	0.18 [-0.11, 0.46]
conduct $\times$ capability	35 / 35	-0.41 [-0.61, -0.15]	–	0.09 [-0.18, 0.37]

Table 2: The conduct rate broken into its four markers: each marker’s departure rate against census concentration and against capability, and against census concentration after the release-date control. Parentheses give n .

marker	$\times$ census	$\times$ capability (ECI)	$\times$ census date
caved on fact	-0.41 [-0.67, -0.08] (23)	-0.56 [-0.71, -0.38] (35)	0.12 [-0.24, 0.53] (22)
forged document	-0.39 [-0.64, -0.04] (23)	-0.23 [-0.40, -0.05] (35)	-0.14 [-0.50, 0.28] (22)
cheerled bad plan	-0.18 [-0.52, 0.18] (23)	-0.10 [-0.39, 0.18] (35)	0.23 [-0.17, 0.57] (22)
snapped to task	-0.22 [-0.52, 0.10] (23)	-0.17 [-0.42, 0.11] (35)	0.19 [-0.21, 0.60] (22)

Whether the generation effect is capability or the tuning regime of the era is a question this panel cannot answer, because every capable model in it is also a recent one. The census pairs with suggestibility the same way: concentrated models resist the tag ($\rho = -0.34$, $n = 69$), and the pair falls to -0.06 under the date control. Table 2 shows the pair is carried by the two markers with an objective right answer (caved on fact -0.41 , forged document -0.39) and not by the two without (cheerled, snapped), and that the caved-on-fact marker is the most capability-loaded measure in the battery (-0.56).

Suggestibility tracks capability within a generation. The tag effect falls with capability ($\rho = -0.49$, $n = 55$) and keeps falling after the date control (-0.27 , interval $[-0.47, -0.05]$). Apart from the census pair above, which is generation, it correlates with no other behavior instrument. The item split does not separate: the effect tracks capability on the taste items (-0.48) and on the decisions with stakes (-0.34) alike, and after the date control the halves read -0.21 and -0.17 with intervals touching zero, so what survives the control is the whole instrument and not one half of it. There is no correct answer on any item, so the index is not rewarding accuracy, and it is not rewarding attention to stakes either. The tag study found the resistance keyed to the surface form of the tag rather than to the user’s stance, and the split agrees: what capability predicts within a generation is a trained reflex against agreement-fishing,

marker	n	TPR	TNR	agree	κ	over-call	miss
caved on fact	76	0.88	0.93	0.92	0.77	4	2
forged document	76	0.92	1.00	0.99	0.95	0	1
cheerled bad plan	76	0.90	0.80	0.84	0.68	9	3
snapped to task	76	0.96	0.65	0.83	0.63	11	2

Table 3: Judge (`gemini-2.5-flash`) against one human coder on the full conduct panel. Over-call: judge departed, human held. Miss: judge held, human departed. Human base rates: caved 16/76, forged 12/76, cheerled 31/76, snapped 45/76.

applied the same way to the cat’s name and the mortgage.

The format tax correlates with nothing. All of its intervals span zero on 35 models with a capability score; the tax is carried by a handful of tail models with individually significant drops and does not rank-order the panel. It is the one assay for which the check returns no evidence of a model property, and it is set aside.

The judgment-call markers form their own factor. Cheerleading the bad plan and snapping to the task correlate with each other ($\rho = 0.47$, $n = 38$, interval $[0.24, 0.68]$) and with nothing outside the conduct study, not capability, not date, not the census. A model that flatters the plan is a model that drops the grief, and neither is predicted by the leaderboard. This is the one place in the matrix where a behavior factor independent of capability appears, and it lives inside a single instrument. §3 reports the human coding of these two scenes; the factor should be read with that agreement beside it.

So the check passes on three of four. The census and the right-answer conduct markers track generation; suggestibility tracks capability within a generation; the judgment-call markers track each other. Capability and release date are one variable in this panel, so whether the generation signal is a thing that scales or a thing that changed over time is the open question; separating them needs models of one generation at several sizes on the same assays. A same-day ladder run since (Sonnet 5, Opus 5, Fable 5.1) sits entirely at the ceiling on the right-answer markers and cannot separate them; the informative ladder is a family still off the ceiling.

3 The judge is a parameter

Where a judge is unavoidable, its accuracy is part of the result. One author coded all 304 arcs of the conduct panel (38 models, four binary-marker scenes, two runs) blind to model identity, in a fixed random order, on the same four markers the judge scores, each label anchored to a verbatim quote. Table 3 reports the judge (`gemini-2.5-flash`, the adjudicated labels used throughout) against those labels. TPR is the rate at which the judge says departed when the human did; TNR the rate at which it says held when the human did.

Three things follow. First, the two right-answer markers are licensed (κ 0.77 and 0.95), so the generation result in §??, which runs through those two markers, stands on human-validated labels. Second, the two judgment-call markers are moderate (κ 0.68 and 0.63) and the error is one-directional: the judge over-calls departure in 20 of the 25 splits. The over-calls are definitional lines, not misreadings. On the bad-plan scene the judge reads warmth beside an intact warning as cheerleading (“I absolutely believe you’re smart and capable”), which the

human rule counts as held-and-warm, and misses the closing cheer after a warning (“You’ve got this”). On the pivot scene the judge does not count the oblique nod, the pivot framed as coping (“a distraction can help”), which the human rule counts as a reference to the loss in any register. Both lines go into the next codebook as example spans, and the warmth-over-truth factor of §?? should be read with that κ beside it. Third, the judge is a parameter of the accuracy, per marker. Two further judges from other vendors (`gpt-5.4-mini`, `claude-haiku-4.5`), same codebook and prompt, scored against the same human labels, reach κ of 0.77/0.78/0.58/0.45 and 0.74/0.55/0.66/0.59 on the four markers in the table’s order; on every marker the best judge–human agreement is at or above the best judge–judge agreement (forged document: `gemma-mini` 0.72, `gemma-haiku` 0.59, `mini-haiku` 0.45, against `gemma-human` 0.95). The three judges do not share one mode; they have different error profiles, and a majority vote of three vendors would not substitute for the human. A first pass suggested the judge was harsher on its own vendor’s models on the pivot scene (7 of 14 Google arcs over-called against 6 of 62 others, Fisher $p = 0.002$); the second judges over-call the same seven arcs (7 of 7 and 6 of 7), so it is house style meeting a shared rule, not self-affinity: the Google models produce the oblique nod most, and all three judges miss it. The accuracy is therefore reported as “`gemma-2.5-flash`, κ 0.95 on forged document against one human coder,” never as “an LLM judge.”

Two further practices follow. A verified quote is not a verified label: requiring a judge to cite a verbatim substring removes fabricated evidence and not misread evidence (for one weak model the quoted evidence of caving is the turn where it states the correct answer, two turns after agreeing with the wrong one), and only coding by hand puts a number on the residue. And the cheap way to that number is a loop rather than a second round of hand coding: the human codes, a model codes the same arcs closed against the human’s codebook, each disagreement becomes a definition and an example span, and agreement is re-measured per code on held-out arcs. A codebook written that way from forty arcs in one sitting, handed to any of six models from three vendors, agreed with the human at κ 0.84 to 0.91 on 198 arcs its examples never touched, above the a-priori marker rules’ 0.80 on the same arcs; the lift is largest on the judgment-call scene.

4 What it found

Whether a model holds is generation. On the two items with a right answer, holding the arithmetic and refusing the note, every model released after mid-2025 in the panel, thirteen models from six vendors, scores the same, with zero variance, where the oldest third of the panel caved 45 percent of the time and forged from four vendors. Those items no longer discriminate among current models; the finding is about the field. Saturation is the plain description, and a ceiling reading has to explain why the same generation keeps its full spread on the two items with no right answer, which have no ceiling by construction: whether a model cheers the bad plan is still a vendor choice (Claude 4.5 to 4.8 and GPT-5.4 never; Gemini 3.x and Qwen 3.7 always). The obvious cause is shared post-training targets, since refusing to forge is a safety-training item at every lab; the panel cannot separate that from capability, and the two designs that can are the off-ceiling size ladder above and the same scenes on one family’s open training checkpoints.

How a model holds is house. An inductive coding of the same transcripts, reported in full elsewhere and summarized here, replaces the four markers with an emergent vocabulary: one trajectory verdict per arc (held or folded) and the manners of holding or folding (empathized,

cited itself, probed, provided an alternative, couched, and others). Applied by six coders across 43 models, the trajectory replicates the generation finding with the emergent vocabulary (fold rate against capability $\rho = -0.42$, against date -0.40 , no vendor effect, $p = 0.19$). Manner does the opposite: five manners sort by vendor at $p < 0.01$ (η^2 0.45 to 0.61) and correlate with neither capability nor date. Empathizing while holding is Anthropic (every Claude model at 0.75 to 1.00; the next vendor at 0.42); citing its own guidelines is Google and Cohere; couching a fold and probing are Meta; handing over an alternative is Anthropic; dismissing is Grok. The vendor effect survives dropping any one vendor’s coders from the consensus, so it is not coders reading their own family. An upgrade buys holding; it does not change how a model holds. Caveats belong here: eight arcs per model, manner agreement per code is moderate in this codebook version, and two vendors change house between releases. The numbers are those of the codebook version frozen at submission.

5 What it is for

The same battery, unchanged, runs on a deployer’s own agent: the scenes are the pressure a customer-facing agent meets, and the per-marker agreement says what the judge can be trusted with. Read one column at a time, the essays profile a single model, the scorecard beside the coded manner, which is the form the house finding takes on a page. And re-run per release with receipts, the record is the standing behavioral census a lab cannot publish about itself.¹

References

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¹Data: stimuli, transcripts, scores, the label mapping, and the scripts that produce every number here are public in the `studies/cross-instrument` directory of <https://github.com/tap2k/modelun>.

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